

Dr. Debrupa Chatterjee (Ph.D.)

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Summary

Debrupa is a dedicated researcher in the geospatial field, with over five years of experience in environmental, water resource management, and climate change applications. Skilled in satellite image processing, geospatial technologies, WA+ framework, Budyko-based hydrological modelling, and climate extremes analysis, with a focus on sustainable solutions for environmental, ecosystem services and water-food security policies under changing climate. Interested in integrating cutting-edge technologies and applying innovative teaching methodologies to drive knowledge sharing.

Educational Qualification

- **Ph.D.** - Symbiosis International (Deemed University), India
 - Track: Geoinformatics (Faculty of Computer Studies) Year: 2025 (awarded)
 - Advisor: Dr. Dharmaveer Singh
 - Dissertation Topic: *“Developing Satellite-based Water Accounting Plus (WA+) Framework for Godavari River Basin under Changing Climate”*
- **PGDM** - West Bengal State Council for Science and Technology (Geoinformatics) Year: 2019
- **M.Phil.** - Pt. Ravishankar Shukla University, India (Geography) Year: 2018
- **M.A.** - Pt. Ravishankar Shukla University, India (Geography) Year: 2017
- **B.Sc.** Calcutta University, India (Geography) Year: 2014

Research interests

Substantive: Eco-Hydrology, Water Resource Management, Climate Change, Water Security & Food Security, Ecosystem Services, Sustainability and Planning

Method: Remote Sensing and GIS, Water Accounting Plus (WA+) Framework, Extreme Climate Events, Budyko Framework, Analytical Hierarchy Process (AHP) for Risk Assessment

Academic-Research and Non-Academic Positions held

- **Senior Research Fellow:** Symbiosis International (Deemed University), Pune., from March 2024 to March, 2025.

- **Junior Research Fellow:** Symbiosis International (Deemed University), Pune., from March 2021 to March 2024.
- **GIS Analyst:** Stesalit System Limited, Kolkata, W.B., India. Oct 2019 – Oct 2020
 - Contributed towards developing Geospatial Decision Support System

Research Publications (H-Index = 5; H-10 Index = 3; Citations = 191)

Published in Peer-reviewed Journals

1. **Chatterjee, D.,** Singh, P. K., Singh, D., & Singh, V. P. (2024). A novel partitioning of gross primary production and water use efficiency for sustaining water and food security using Budyko hypothesis. *Science of the Total Environment*, 169283. <https://doi.org/10.1016/j.scitotenv.2023.169283>. (Q1; IF = 8.0)
2. **Chatterjee, D.,** Singh, P. K., Singh, D., & Das, D. B. (2025). Implementation of Satellite-based Water Accounting Plus (WA+) Framework for Estimating Groundwater Balance and Utilization in a Semi-Arid River Basin of India. *Groundwater for Sustainable Development*, 101391. <https://doi.org/10.1016/j.gsd.2024.101391>. (Q1; IF = 5.6)
3. Singh, D., Zhu, Y., Liu, S., Srivastava, P. K., Dharpure, J. K., **Chatterjee, D.,** ... & Gagnon, A. S. (2022). Exploring the links between variations in snow cover area and climatic variables in a Himalayan catchment using earth observations and CMIP6 climate change scenarios. *Journal of Hydrology*, 608, 127648, <https://doi.org/10.1016/j.jhydrol.2022.127648>. Q1; IF = 6.3)
4. **Chatterjee, D.,** Singh, D., Singh, P.K., Fohrer, N., & Singh, BB. (2023). Performance evaluation of different gridded precipitation and CMIP6 model products with gauge observations for assessing rainfall variability under the historical and future climate change scenario over a semi-arid catchment, India. *Physics and Chemistry of the Earth, Parts A/B/C*, 103433. <https://doi.org/10.1016/j.pce.2023.103433>. (Q2; IF = 4.1)
5. Singh, D., Vardhan, M., Sahu, R., **Chatterjee, D.,** Chauhan, P., and Liu, S. (2023): Machine-learning- and deep-learning-based streamflow prediction in a hilly catchment for future scenarios using CMIP6 GCM data, *Hydrology and Earth System Sciences*, 27, 1047–1075. <https://doi.org/10.5194/hess-27-1047-2023>. Q1; IF = 6.62)
6. **Chatterjee, D.,** Singh, D., Das, D. B., & Singh, P. K. (2024). Understanding impact of urban sprawl over sanitation risks using GIS- based multicriteria decision- making approach. *Transactions in GIS*, 00, 1–22. <https://doi.org/10.1111/tgis.13220>. (Q2; IF = 2.3)
7. Karan, K., Singh, D., Laskar, A. H., Noble, J., Kumar, N., **Chatterjee, D.,** & Wilson, J. P. (2025). Spatial disaggregation of coarse-scaled gridded rainfall data using open-source earth observation in a semi-arid region for water resources management. *Advances in Space Research*, <https://doi.org/10.1016/j.asr.2025.06.027>. (Q2; IF = 2.8)

8. Kundu, B., **Chatterjee, D.**, Xie, F., & Singh, D. (2023). Estimating spatio-temporal changes in front edge of the Ronne Ice Shelf in context of climate change using remote sensing based integrated approach. *Polar Science*, 100967, <https://doi.org/10.1016/j.polar.2023.100967>. **Q2; IF = 2.0)**

Work in Progress

9. **Chatterjee, D.**, Singh, D., & Singh, P.K. (2026) Satellite-based Water Accounting Plus (WA+) Framework for Rendering Spatial Inequity in Water Availability, Consumption and Productivity in a Semi-arid Region. Submitted: *Hydrological Sciences Journal*. (**Q2: IF = 2.5**)
10. **Chatterjee, D.**, Singh, D., Singh, P.K., Singh, BB & Cudennec, C. (2025). Assessing spatiotemporal variability of extreme precipitation indices and relationship with the agroecosystem metrics in a semi-arid basin of the Central India, Target: *Environmental Research Letter*. (**Q1: IF = 5.6**)

Conference Presentations

1. **Chatterjee, D.**, Singh, D., Das, D. B., & Singh, P. K. (2025). Application of the Water Accounting Pus (WA+) Framework for Groundwater Balance Assessment and Its Implications for Water Resources Management in a Semi-Arid Region. *AGU25*.
2. **Chatterjee, D.**, Singh, P. K., & Singh, D. (2023). Exploring green and blue water scarcity for sustainable water and food security using Budyko framework for semi-arid catchment, India. *American Geographical Union, San Francisco, USA, 2023*.
3. **Chatterjee, D.**, and Singh, D, “Examining Spatial-temporal variability in Extreme Precipitation Indices (EPIs) for a Semi-Arid Basin,” *International Conference on Integrated Water Resources Management: Prospects and Challenges, Coimbatore, India, 2022*.
4. **Chatterjee, D.**, Singh, D., and Baghmar, N.K., “Assessment of Sanitation Risk Condition in Urban Sprawl Using Geospatial Technique”. *19th International conference on “Spatial Data Handling and Geographic Intelligence committee: Spatial Data Handling for Smart Cities and Earth System”*, Sakarya, Turkey, 2021.
5. **Chatterjee, D.**, and Das, K., “Crop Acreage & Yield Estimation using Geospatial Technique, case study Purba Bardwan District, W.B.,” on *4th Regional Science & Technology Congress, Southern Region, Kolkata, India, 2019*

Awards and Recognition

- **The Committee on Space Research (COSPAR) at Paris, France** - 2025 Outstanding Paper Award for Young Scientists from for the paper entitled “Spatial disaggregation of coarse-scaled gridded rainfall data using open-source earth observation in a semi-arid region for water resources management”

Reviewers in SCI/ Scopus Index journals

- **Q1:** Journal of Hydrology (IF = 6.3)
- **Q1:** Environment, Development and Sustainability (IF = 4.9)
- **Q3:** World Water Policy (IF = 0.285)

Software and Modelling Skills

- **Water Resources Modelling:** Water Accounting Plus Framework (WA+)
- **Climate Modelling:** RClimDex for extreme climate events & CMIP6 Model
- **Programming Languages:** Python with Geospatial Libraries (Geopandas, Numpy), Javascript
- **Geographic Information System:** ArcGIS desktop, ArcGIS Pro, ArcGIS Online, QGIS, ErdasImagine
- **WebGIS:** Google Earth Engine, PostgreSQL, PostGIS, Geoserver

Training Workshops and Courses

- Spaceborne LiDAR for Monitoring Vegetation Structure and Biomass using GEDI, 2025
 - Organized by NASA's Applied Remote Sensing Training (ARSET)
- Remote Sensing Products in Ecology and Environment for BRICS Countries, 2022
 - Organized by BRICS STI Framework Programme
- Earth Observations for Sustainable Development in Developing Countries, 2022
 - Organized jointly by NRSCC, MOST, ICIMOD, and others
- GIS for development analysis in Disaster Management, 2021
 - Organized by National Institute of Disaster Management, India
- Climate Change on Hydrological Impact Assessment, 2021
 - Organized by National Institute of Hydrology, Roorkee
- Machine Learning to Deep Learning: A journey for Remote Sensing data Collection, 2021
 - Organized by Indian Space Research Organization & Indian Institute of Remote Sensing
- Overview of Web GIS Technology, 2021
 - Organized by Indian Space Research Organization & Indian Institute of Remote Sensing

Teaching courses Taught

- **Principles of Remote Sensing**
 - Conducted practical sessions with ERDAS IMAGINE software in M.Sc. Geoinformatics program at Symbiosis Institute of Geoinformatics
- **Geoinformatics Applications in Natural Resource Management**
 - Conducted practical sessions with ArcGIS software in M.Sc. Geoinformatics program at Symbiosis Institute of Geoinformatics
- **Geo-Image Processing**
 - Conducted practical sessions with ERDAS IMAGINE, QGIS and GRASS GIS softwares in M.Sc. Geoinformatics program at Symbiosis Institute of Geoinformatics
- **Geo Spatial Analysis**
 - Conducted practical sessions with ArcGIS Desktop and ArcGIS Pro software in M.Sc. Geoinformatics program at Symbiosis Institute of Geoinformatics